

ME594: 3D Robotics, HW 5

Homework due on 02-16-2025, Topics: Inverse Kinematics.

1. **Inverse Kinematics of A1 Leg:** The inverse kinematics is normally computed using numerical methods (e.g., `fsolve`). But this adds to the computation time. For certain robot geometries, the inverse may be computed analytically. This is highly desirable as it saves computational time, especially for real-time control. Here, we will compute the analytical inverse for the Unitree A1 quadruped leg.

The origin of the frame is shown as a white sphere. The axes are shown (red, green, and blue are x-, y-, and z-axis respectively). The reference position of the red square from the origin is: x_{ref} , y_{ref} , z_{ref} and the thigh and shank length $L = 0.2$ m. Compute the abduction angles q_a , the hip angle q_h , and the knee angle q_k so that the foot (yellow sphere) is on the reference (red square). Download and unzip the file `unitree-a1-leg.zip`. I have provided `mj_a1_inverse_kinematics_leg.py`, which you will run. I have also provided you with the `scene.xml` and `a1_leg.xml`, but you must download the mesh files here to get the .py file up and running. https://github.com/google-deepmind/mujoco_menagerie/tree/main/unitree_a1.

Once you compute the formulae, do the two things below

- (a) In the function `inverse_kinematics_analytic.py`, substitute the values of q_a , q_h , and q_k (search HINT1). Then run `mj_a1_inverse_kinematics_leg.py` to see if the foot is on the red square.
- (b) In the function `mj_a1_inverse_kinematics_leg.py` try different reference values (search HINT2).

HINTS: From the figure $\ell = \sqrt{\ell_x^2 + \ell_y^2 + \ell_z^2}$. Compute q_a by using trigonometry in the y-z (green-blue) plane. Compute q_h and q_k by using trigonometry in the x-z (red-blue) plane.

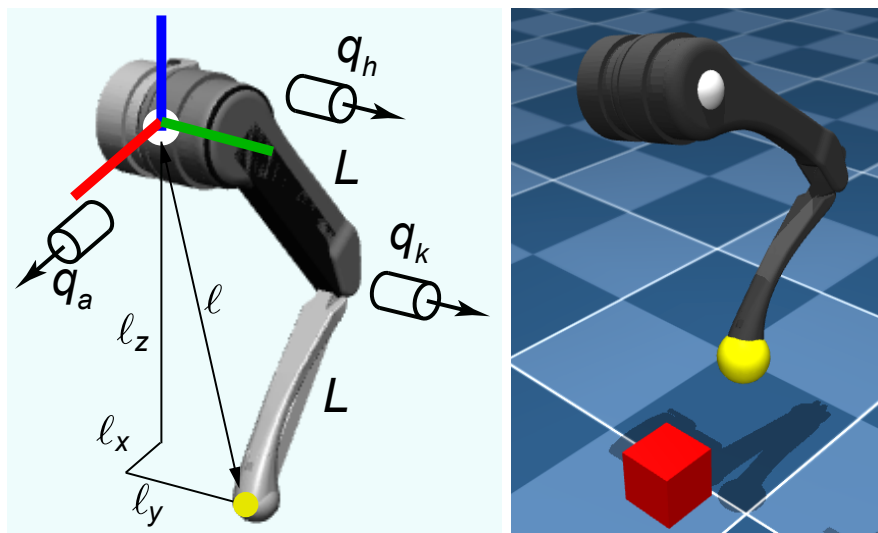


Figure 1: **Unitree A1 leg:** (a) Dimensions for inverse kinematics (b) MuJoCo scene. Compute the joint angles q_a , q_h , and q_k such that the foot (yellow sphere) is on the reference (red square).

Submission:

- **Gradescope:** None.
- **Email:** Rename the folder and zip it: *unitree_a1_leg-names.zip* where *names* should be replaced by names of the team member(s) and sent to pranav@uic.edu. Please add “HW5 A1 leg” to your subject line. If submitting as a group, please copy your group member. You can remove any mesh files from your submission because they might be too big to attach to the email.

2. **Inverse Kinematics of Trossen Robotics WidowX 250 S:** Figure 2 shows the WidowX 250S by Trossen Robotics. A video of the robot in action is here: <https://www.youtube.com/watch?v=l.10IH0JNxI>. I have provided the wx250s.xml files, the remaining scene and meshes can be downloaded from github https://github.com/google-deepmind/mujoco_menagerie/blob/main/trossen_wx250s/wx250s.xml Write code to draw a circle in the vertical plane (y-z) while maintaining the x-position end-effector orientation (e.g., quaternion $[1, 0, 0, 0]$) to a constant value. Your code should show an animation followed by a plot of the reference circle and the trajectory produced by the end-effector.

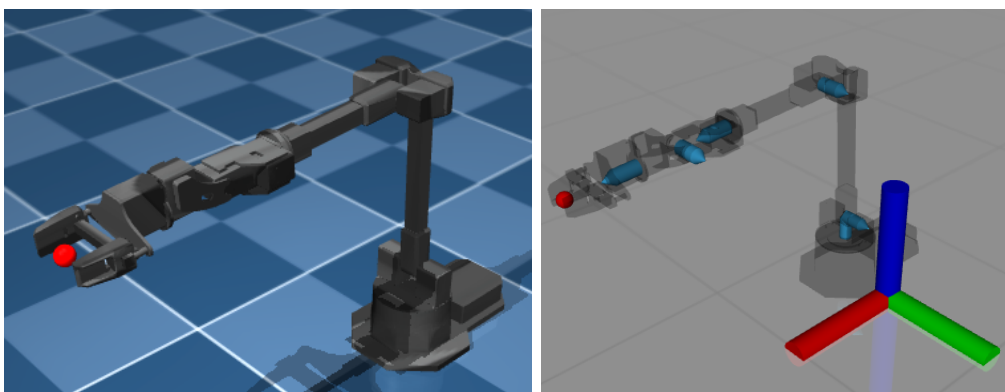


Figure 2: Perspective view of the Trossen WidowX 250 S: The robot has 6 joints as shown on the figure on the right. The world frame is at the base of the WidowX but moved to the side in the figure to avoid clutter. The body frames are in the same direction as the world frame and centered at the joints.

Submission:

- **Gradescope:** None.
- **Email:** Submit a zipped folder: *w250s_ik-names.zip* where *names* should be replaced by names of the team member(s) and sent to pranav@uic.edu. Please add “HW5 wx250s” to your subject line. If submitting as a group, please copy your group member. You can remove any mesh files from your submission because they might be too big to attach to the email.